Yash Saxena

PhD Student, Computer Science, UMBC | Trustworthy AI, RAG & Neurosymbolic AI
Advisor: Dr. Manas Gaur | KAI^2 Lab

ysaxena1@umbc.edu | yashsaxena21.github.io/Portfolio/ | YashSaxena21 | yash-saxena-a18b251bb | G Scholar

Research Focus

I study trustworthy RAG through **Interpretable Knowledge Flow (IKnowFlow)**, making the evidence trail from retrieved sources to generated claims explicit across high-stakes domains such as law, finance, and scientific research.

Skills

Research Areas	Information Retrieval, Large Language Models, Neurosymbolic AI, Trustworthy AI, NLP
Methods	Preference Optimization (DPO, PPO), Retrieval Modeling, Knowledge Graphs, Evaluation Design
Systems	Python, PyTorch, Hugging Face, FAISS, Slurm, LangChain, LlamaIndex

Education

University of Maryland, Baltimore County — *Baltimore, Maryland, USA*
PhD, Computer Science — *Jan 2025 - Present*

- Research Assistant | Advisor: Dr. Manas Gaur | GPA: 4.0/4.0

Professional Experience

TIB, Leibniz Information Centre for Science and Technology — *Hannover, Germany*
Visiting Research Intern, Hosted by Prof. Sören Auer — *May 2026 – Aug 2026*

- Built **Dynamic Cognitive Knowledge Graphs (DCKG)** on **ORKG** scholarly knowledge for **TIB AIssistant** research workflows.

Knowledge-Infused AI & Inference (KAI2) Lab, UMBC — *Baltimore, MD, USA*
PhD Research Assistant, Advised by Dr. Manas Gaur — *Jan 2025 – May 2026*

- Developed **IMRNNs**, interpretable embedding-modulation adapters for dense retrieval, improving recall by **+7.14% across seven benchmarks**.
- Developed **METEORA** for rationale-driven evidence selection in RAG, improving answer accuracy by **+33.34%** with **50% fewer evidence chunks**.
- Compared generation-time and post-hoc citation; post-hoc reached **78%** human-evaluated correctness vs **69%** for generation-time citation.

Knowledge-Infused AI & Inference (KAI2) Lab, UMBC — *Baltimore, MD, USA*
Remote Research Intern, Advised by Dr. Manas Gaur — *Aug 2024 – Jan 2025*

- Fine-tuned **LLaMA-3.1 70B** with **Direct Preference Optimization (DPO)** and integrated it into a RAG pipeline for evidence-grounded answering.
- Designed ablations and evaluation protocols and contributed to the **REASONS** benchmark, accepted at **IEEE DSAA 2026**, through an IARPA-supported project.

Artificial Intelligence Institute, University of South Carolina (AIISC) *Columbia, SC, USA*

REMOTE RESEARCH INTERN, ADVISED BY DR. AMIT SHETH *Aug 2023 – Jul 2024*

- **Project: LLM Applications and Dataset Construction.** Developed LLM-based components using **LangChain** and **LlamaIndex** for research prototypes and experiments.
- Built web-scraping workflows to construct structured datasets for downstream modeling experiments.

Publications

Yash Saxena, Ankur Padia, Mandar Chaudhary, Kalpa Gunaratna, Srinivasan Parthasarathy, Manas Gaur. "Ranking Free RAG: Replacing Re-ranking with Selection in RAG for Sensitive Domains". **ICML 2026**. Link

Deepa Tilwani*, **Yash Saxena***, Seyedali Mohammadi, Ankur Padia, Edward Raff, Amit Sheth, Srinivasan Parthasarathy, Manas Gaur. "Abstention vs. Hallucination: Benchmarking LLM Source Attribution for Sentence-level Scientific Citations". **IEEE DSAA 2026, Dataset & Benchmark Track (Oral)**. * Equal contribution.

Yash Saxena, Ankur Padia, Kalpa Gunaratna, Manas Gaur. "IMRNNs: An Efficient Method for Interpretable Dense Retrieval via Embedding Modulation". **Findings of EACL 2026.** Link

Yash Saxena, Manas Gaur. "Neurosymbolic Retrievers for Retrieval-Augmented Generation". **IEEE Intelligent Systems 2026.** Link

Yash Saxena, Raviteja Bommireddy, Ankur Padia, Manas Gaur. "Generation-Time vs. Post-hoc Citation: A Holistic Evaluation of LLM Attribution". **NeurIPS 2025 LLM Evaluation Workshop**. Link

Seyedreza Mohseni, Seyedali Mohammadi, Deepa Tilwani, **Yash Saxena**, Gerald Ketu Ndawula, Sriram Vema, Edward Raff, Manas Gaur. "Can LLMs Obfuscate Code? A Systematic Analysis of Large Language Models into Assembly Code Obfuscation". **Proceedings of the AAAI Conference on Artificial Intelligence 2025**, 39(23), 24893-24901. Link

Presentations

August 2026. Doctoral Consortium presentation: "IKnowFlow: Trustworthy RAG for Sensitive Domains". **IJCAI-ECAI 2026 Doctoral Consortium, Bremen, Germany.**

July 2026. Poster: "Ranking Free RAG: Replacing Re-ranking with Selection in RAG for Sensitive Domains (METEORA)". **ICML 2026, Seoul, South Korea.**

December 2025. Poster: "Generation-Time vs. Post-hoc Citation: A Holistic Evaluation of LLM Attribution". **NeurIPS 2025 LLM Evaluation Workshop, San Diego, CA, USA.**

June 2025. Talk: "RASOR: Contextual Legal Intelligence via Rationalized Selection & Refinement in RAG". **Bloomberg Law, Language, and AI Symposium, New York, NY, USA.**

Mentoring

Aug–Nov 2025. Mentored **Raviteja Bommireddy** (B.Tech CSE, IIITDM Kancheepuram) on LLM attribution and citation evaluation; co-authored a NeurIPS 2025 workshop paper.

May 2026–Present. Mentoring **Shaunjit Chakrabarti** (B.Tech CSE, Manipal Institute of Technology Jaipur) on AI agents for legal-case simulation and reasoning.

Recognitions, Fellowships, & Grants

RECOGNITIONS

2026	**Summer research scholarship**, TIB.
2026	**Travel grant recipient**, SIGAI-supported IJCAI-AIJ Doctoral Consortium.
2022	**Winner, UNESCO India–Africa Hackathon**, Ministry of Education (India).
2022	**Winner and team lead, Smart India Hackathon**, Ministry of Education (India).